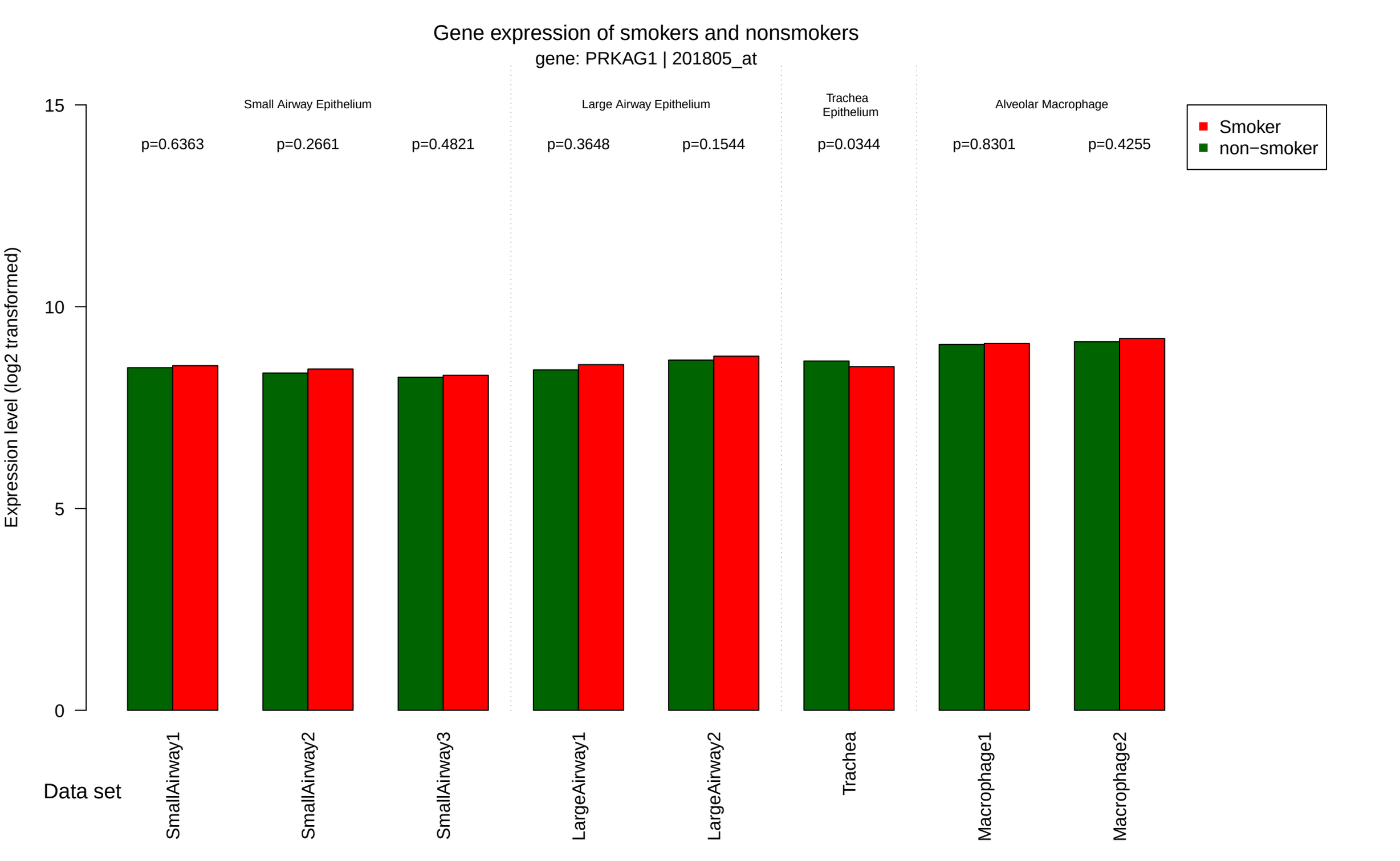
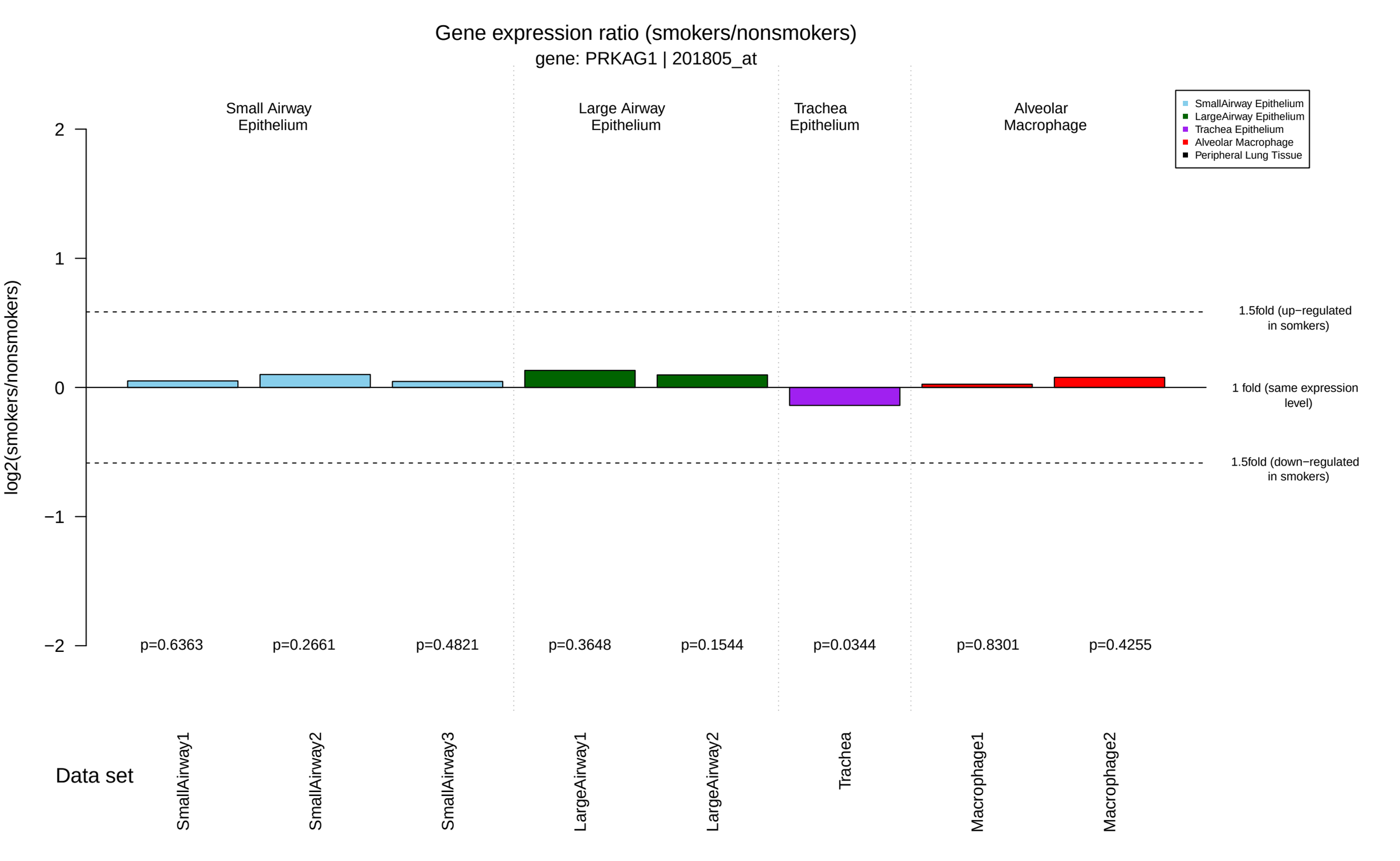
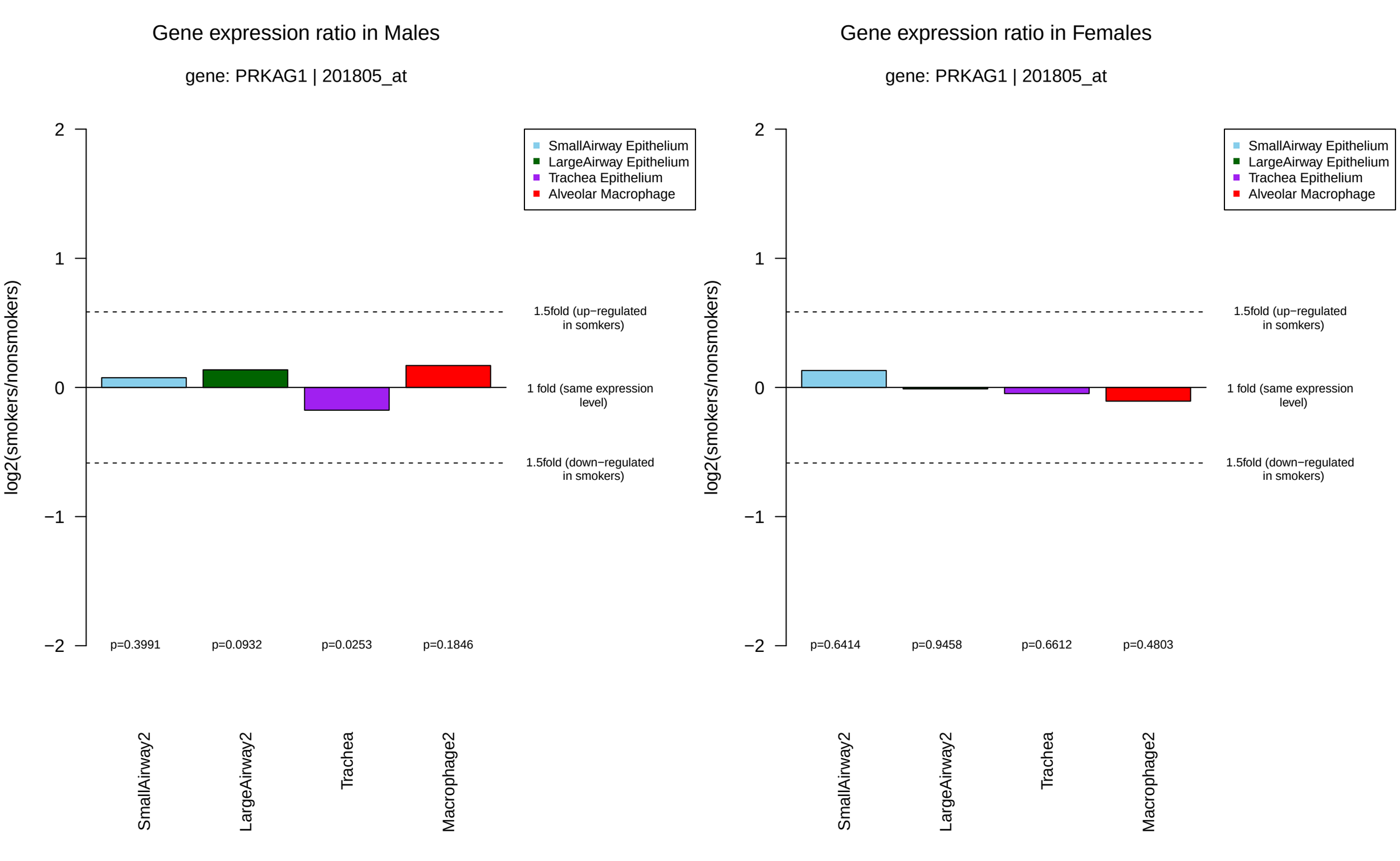
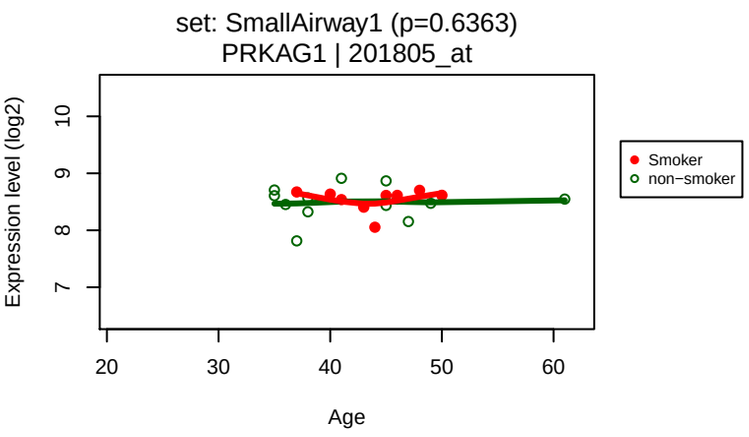
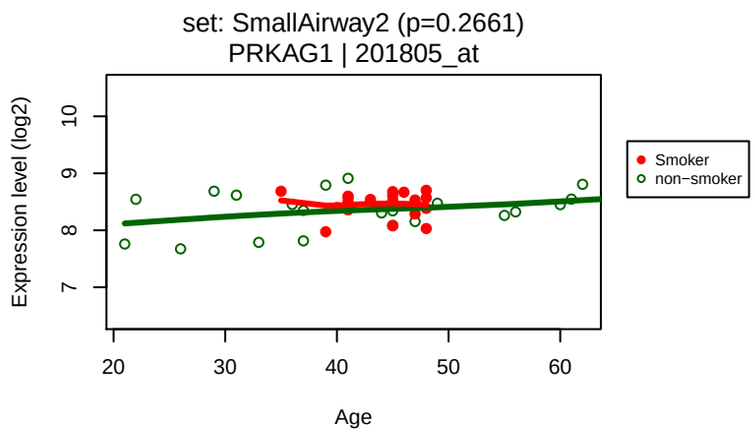
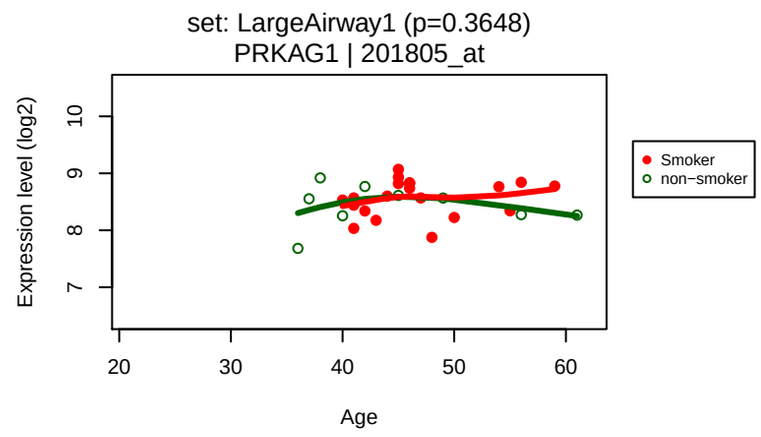

Gene expression of smokers and nonsmokers
gene: PRKAG1 | 201805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6363
p=0.2661
p=0.4821
p=0.3648
p=0.1544
p=0.0344
p=0.8301
p=0.4255
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

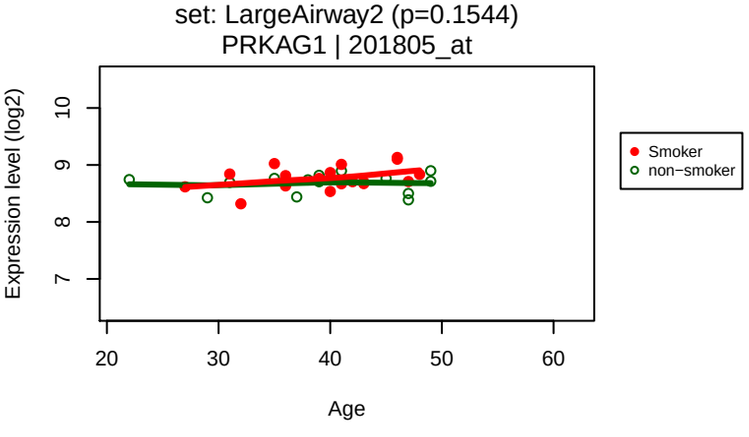

Gene expression ratio (smokers/nonsmokers)
gene: PRKAG1 | 201805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6363
p=0.2661
p=0.4821
p=0.3648
p=0.1544
p=0.0344
p=0.8301
p=0.4255
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

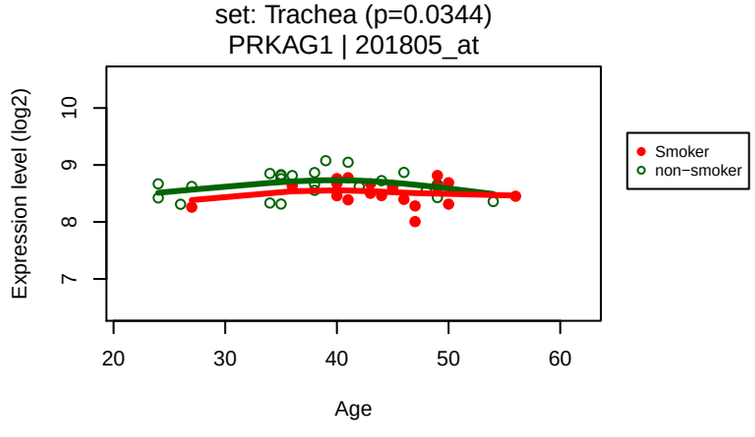

Gene expression ratio in Males
gene: PRKAG1 | 201805_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3991
p=0.0932
p=0.0253
p=0.1846
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAG1 | 201805_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6414
p=0.9458
p=0.6612
p=0.4803
SmallAirway2
LargeAirway2
Trachea
Macrophage2

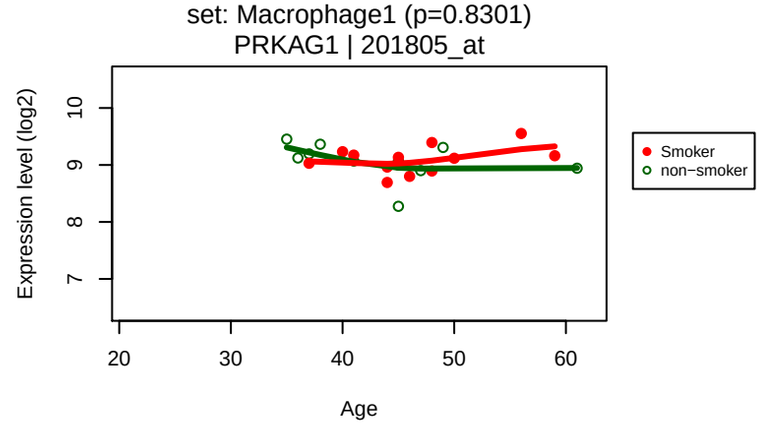

set: SmallAirway1 (p=0.6363)
PRKAG1 | 201805_at
Expression level (log2)
Age
Smoker
non−smoker

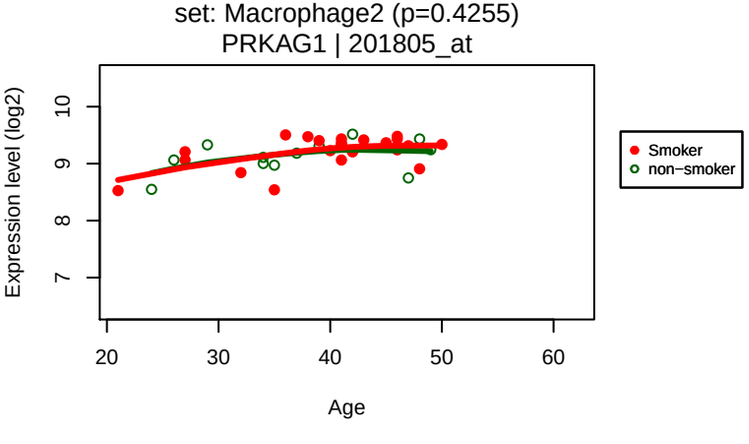

set: SmallAirway2 (p=0.2661)
PRKAG1 | 201805_at
Expression level (log2)
Age
Smoker
non−smoker

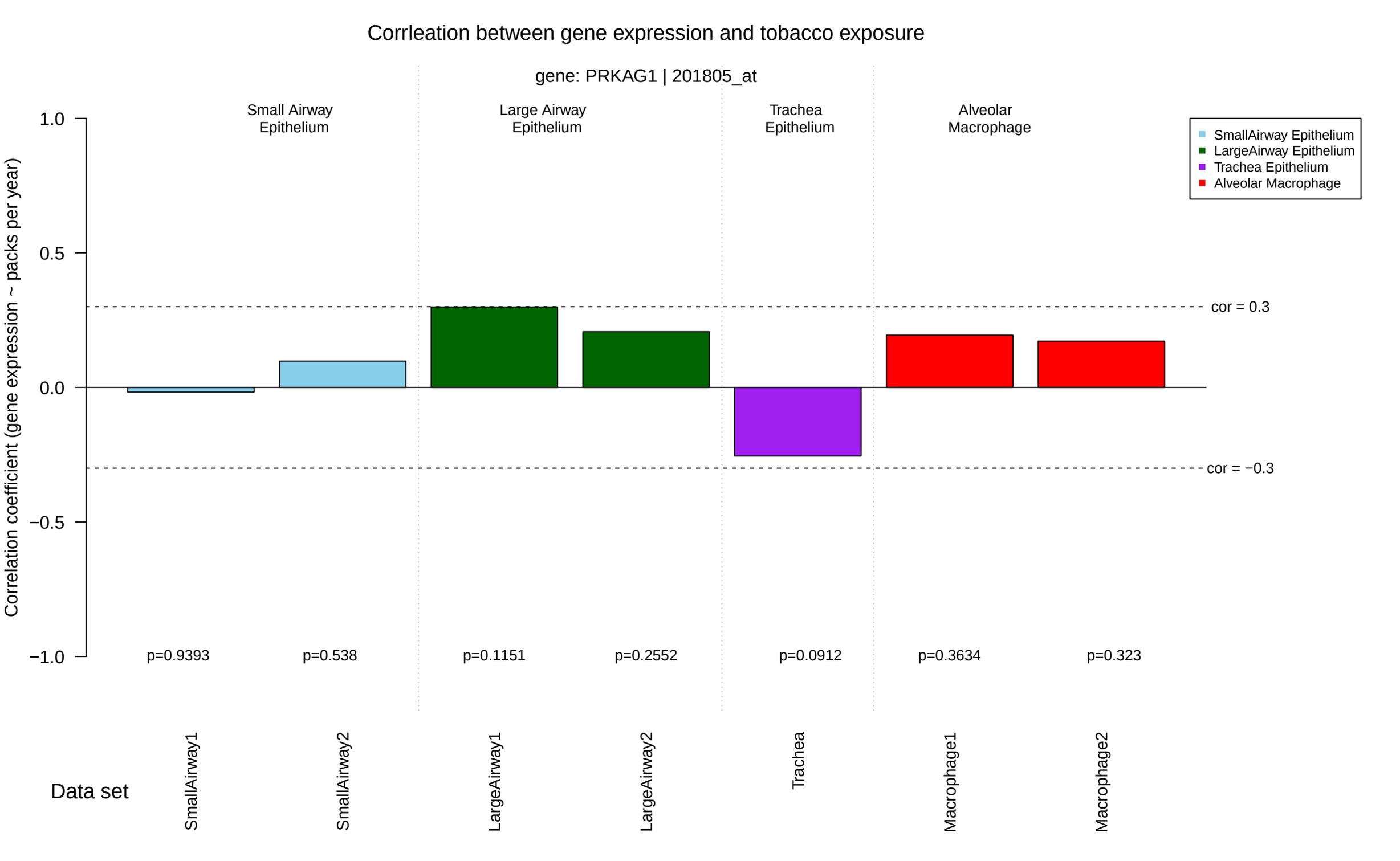

set: LargeAirway1 (p=0.3648)
PRKAG1 | 201805_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1544)
PRKAG1 | 201805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0344)
PRKAG1 | 201805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8301)
PRKAG1 | 201805_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4255)
PRKAG1 | 201805_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAG1 | 201805_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9393
p=0.538
p=0.1151
p=0.2552
p=0.0912
p=0.3634
p=0.323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2